SULIT

FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA
MID-TERM EXAMINATION SEMESTER 1 2022/2023

SUBJECT CODE	**:**	**SECJ 2253**
SUBJECT NAME	**:**	**REQUIREMENTS ENGINEERING & SOFTWARE MODELING**
SECTION	**:**	**01, 02, 03, 04**
TIME	**:**	**8.00 – 9.30 PM**
DATE/DAY	**:**	**23 MAY 2023, TUESDAY**
VENUES	**:**	**DEWAN KULIAH 4 & 5, BLOCK P19**

INSTRUCTIONS:

SECTION A :	**10 MULTIPLE CHOICE QUESTIONS**	**(20 MARKS)**
SECTION B :	**4 STRUCTURED QUESTIONS**	**(80 MARKS)**
TOTAL		**(100 MARKS)**

(Please Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **SIX (6)** printed pages excluding this page.

SECTION A: MULTIPLE CHOICE QUESTION [20 MARKS]

Choose the **BEST** answer for the question. Write the answer in the given booklet.

1. Which of the following is the **BEST** reason related to insufficient and poor requirements engineering?
   A. Unrealistic time frames
   B. Changing requirements specifications
   C. Lack of executive support
   D. New technology

2. Which of the following requirements is **MOST** relevant for the functional requirements?
   A. The system must allow users to verify their accounts using email address within 2 days the email sent to the users.
   B. The system must send a confirmation email within 5 seconds whenever an order is placed.
   C. The system must allow blog visitors to sign up for the newsletter using their email within 2 days from the date they register in the system.
   D. The system must send a confirmation email whenever a booking is done.

3. Which of the following is **BEST** represents the system context object in terms of interface for the residential college management system?
   A. Admin clerk (college officer)
   B. Room booking policies
   C. Personal data privacy
   D. Rental period terms

4. Which of the following statement is **FALSE** corresponding to the grey zone?
   A. Only the context boundary can contain a grey area.
   B. A grey zone in the system context means requirements are incomplete, not decided or information is missing.
   C. Refer to the initial vague separation between system and its context at the beginning of the requirements engineering process
   D. A newly identified external system in the context might become part of the grey zone in the system boundary

5. Which of the following is **NOT** a source that needs to be considered to develop an accounting system for a local Malaysian manufacturing company?
   A. Relevant Malaysia law for import and transport
   B. Procedures for the export of the drugs
   C. Business process of manufacturing industry
   D. Stakeholder needs

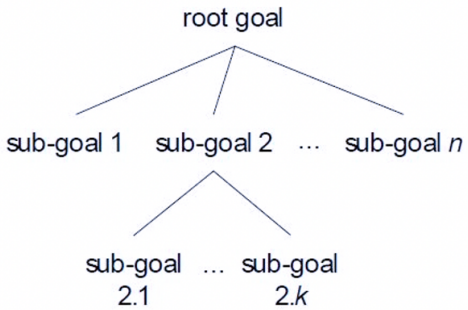
6. Which of the following is a characteristic of the **Basic/Dissatisfier** category in the Kano Model?
   A. Customer satisfaction will be increased due to not fulfilling customer needs
   B. The system must realize this requirement to enable market entry
   C. Customer consciously demands the realization in the system
   D. Customer satisfaction increase disproportionately

7. Which of the following is the **MOST** related risk of a field observation technique during requirements elicitation?
   A. Process of identify the basic factors and detailed requirements
   B. Involve with the observation in the real context
   C. Identifies only performance factors
   D. Missing motivation and feedback

8. Which of the following statements is **NOT** relevant to the goal model?
   A. Can be visualized in the form of diagram.
   B. Can be documented in natural language.
   C. Describes the intention of a requirements engineer.
   D. Has the decomposition relationships between higher and subordinate goals.

9. Which of the following statements is **INCORRECT** in describing the purpose of the swimlane model?
   A. To model the possible use case of the system.
   B. To propose solutions for the to-be system.
   C. To model the functions of an existing system.
   D. To identify possible solutions for the existing system.

10. Which of the following statement is **BEST** to represent the following goal model?



   A. Every sub-goal must be fulfilled so that the root goal is fulfilled.
   B. Super-goal is considered fulfilled even if none of the root goal is fulfilled.
   C. At least two sub-goals must be fulfilled so that the root goal is fulfilled.
   D. It is sufficient if at least one sub-goal is fulfilled so that the root goal is met.

SECTION B: STRUCTURED QUESTION [80 MARKS]

There are 5 questions. The marks for each part of the question are as indicated. Write the answer in the given booklet.

Question 1 [26 marks]

The XYZ company has embarked on developing a new Human Resource (HR) system to automate HR processes, such as recruitment, performance management, and employee development. The new HR system will utilize a centralized database to store and manage employee information, streamlining the tracking of employee performance, identification of skill gaps, and provision of relevant training and development opportunities. Moreover, the system will empower managers to set performance goals for their team members and monitor progress towards achieving those goals. However, during the early requirements engineering process with the consultant, the XYZ company is uncertain about whether to include the payroll module in the new HR system. Integration with the existing payroll system is essential to ensure accurate and timely payment processing. Additionally, the XYZ company needs to decide whether the new HR system should complies with relevant data privacy regulations, such as the General Data Protection Regulation (GDPR). The new HR system is expected to handle sensitive employee data appropriately, with provisions for consent management, data access controls, and secure data storage.

Based on the above XYZ company scenario, answer the following questions (a) until (g):

(a) Describe the following terms: (6 marks)
   i. Requirements engineering
   ii. Stakeholder
   iii. Constraint requirement

(b) Suggest **THREE (3)** different types of requirements [functional | quality | constraints] for the new HR system. (6 marks)

(c) Name **TWO (2)** suitable diagrams that can be used to document the HRS system and XYZ company context. (2 marks)

(d) Define the following terms and give example related to the XYZ company scenario. (4 marks)
   i. System context
   ii. System boundary

(e) Identify and analyse **TWO (2)** potential grey areas that may arise during the requirements engineering process for the XYZ company in the development of their new HR system. (8 marks)

Question 2 **[34 marks]**

As a consultant for ABC company, a large retail chain with a global presence, you have been tasked with developing a new customer service management system that can handle the growing number of customer inquiries and complaints. The system will feature an intelligent virtual assistant to provide quick and efficient solutions to customers. During the first meeting at the main branch, customer service representatives, managers, and the IT department gathered to discuss the project. The meeting covered the sharing of the customer service manual, the company's service level agreement, and customer feedback reports. Additionally, the IT head of department presented an analysis of the existing customer service system, including call logs, chat transcripts, and email exchanges. The predecessor system able to log customer inquiries, assign customer to the appropriate representative, and track communication progress. The customer service representatives highlight that they wish the new system have ability to prioritize urgent inquiries, automate simple tasks, and provide personalized responses based on customer history.

Based on the above ABC company scenario, answer the following questions (a) until (g):

(a) Name the **THREE (3)** type of requirements sources that you can refer during elicitation process. (3 marks)

(b) State the example for each requirement source in (a). (3 marks)

(c) Suggest **FOUR (4)** customers satisfaction based on Kano Model factors towards the new customer service management system (as shown in Table 1). (8 marks)

Table 1

Kano Factor	Customers Satisfaction
i. Basic (Dissatisfier)	
ii. Performance (Satisfier)	
iii. Performance (Satisfier)	
iv. Excitement (Delighter)	

(d) Describe **THREE (3)** elicitation techniques other than creativity based, which suitable for Kano Model basic and performance factors - that could be implemented in the context of XYZ company. (6 marks)

(e) Outline the procedures to implement the brainstorming session as one of the creativities based for elicitation technique. (6 marks)

(f) Justify whether the brainstorming technique help to identify **excitement factor** of Kano Model for the new customer service management system featuring an intelligent virtual assistant. (2 marks)

(g) *Creativity elicitation techniques is suitable to use due to its potential to gain ideas in a short time. However, it is not suitable for complex problems and precise requirements. Prototyping is a supporting technique in requirements elicitation.* Justify how can **throw-away and evolutionary prototyping techniques** be utilized to complement creativity elicitation techniques? (6 marks)

Question 3 **[7 marks]**

JKSNJ agency aims to enhance the efficiency of their file management process. They are planning to implement a new file management system that prioritizes reliable file storage with strong security measures and strict privacy controls, ensuring that only authorized users have access to the system. The following Figure 1 shows the goal model for the proposed file management system.

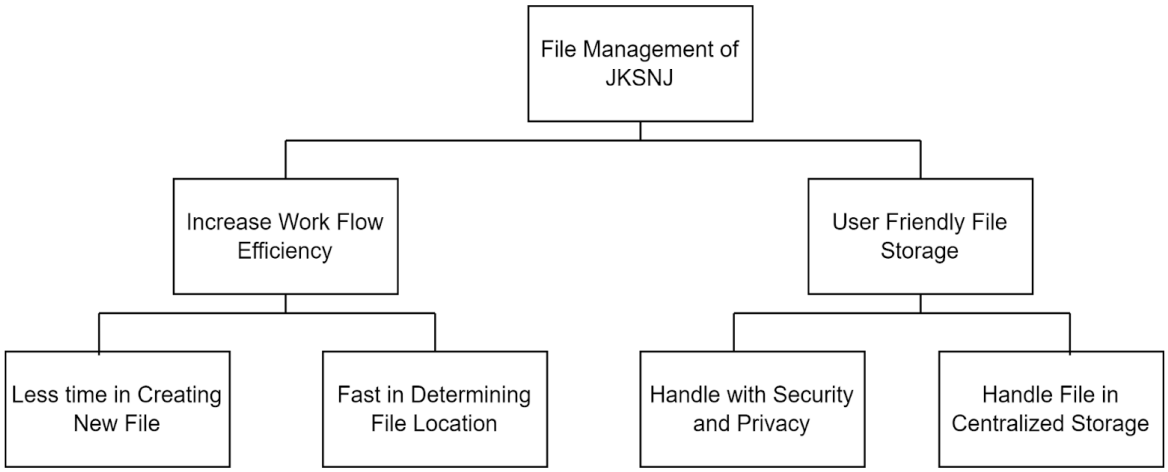


Figure 1: Goal Model of JKSNJ File Management System

Based on the JKSNJ agency scenario, identify **THREE (3)** improvements that should be made to the goal model of the proposed file management system in **Figure 1**.

Question 4 **[13 marks]**

The MBIP agency is exploring the possibility of developing an Online Complaint System (OCS) to simplify the process of gathering feedback from the community regarding health service issues. The objective is to establish a more efficient method for community members to submit complaints directly to MBIP as the local authority. At present, there is a lack of an integrated and centralized system to promptly manage complaints. Additionally, the information obtained from complaints lacks detail and does not effectively convey the level of urgency associated with the raised issues.

Based on the above scenario of OCS, answer the following (a) and (b) questions:

(a) Describe the goal diagram for OCS and its decomposition in **Figure 2**. (5 marks)

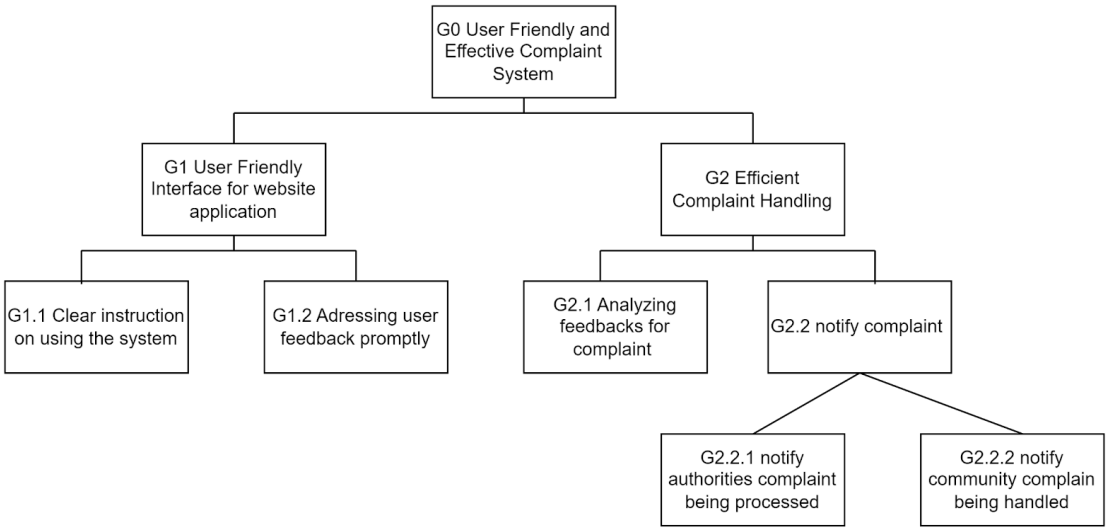


Figure 2: Goal Diagram for OCS

(b) **Figure 3** presents the Swimlane diagram for the overall OCS current As-Is processes. Analyze **FOUR (4)** potential errors or problems in the **Figure 3** Swimlane diagram. (8 marks)

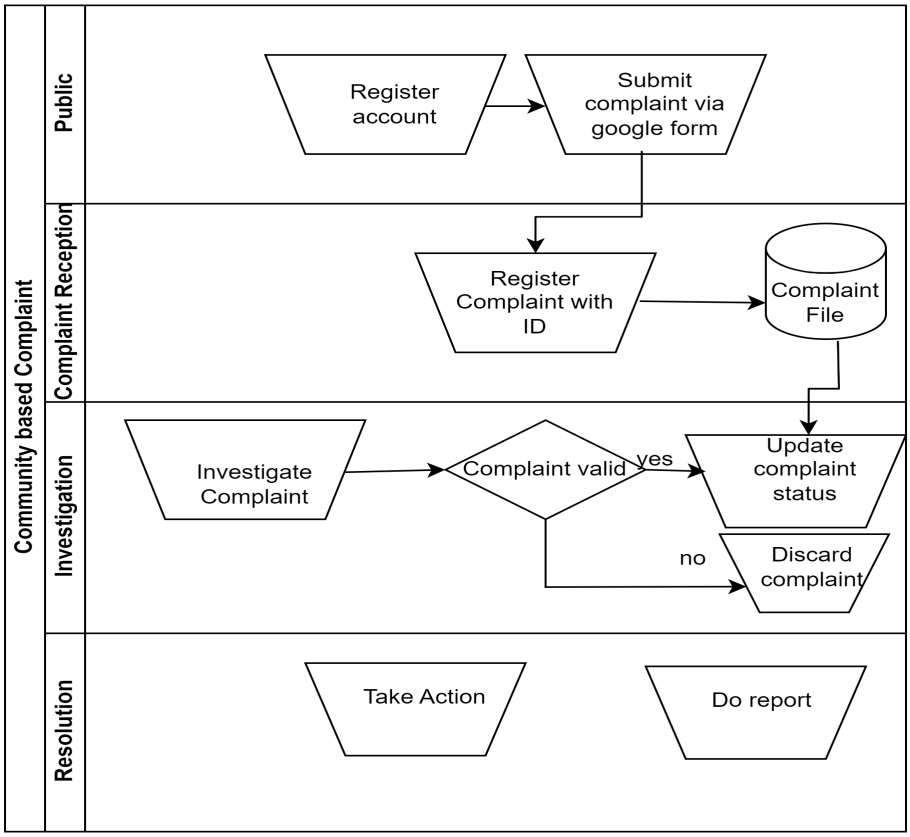


Figure 3: Swimlane Diagram for OCS As-Is Process